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2024

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning
- and/or calculations

Total marks: 100

Section I – 10 marks (pages 2-7)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 8-33)

- Attempt Questions 11-34
- Allow about 2 hours and 45 minutes for this section

Directions to School or College

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Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

1. A marketing team is investigating the relationship between fuel consumption (km/L) and a car's weight (kg). Data has been collected and a least squares regression line is used to predict this relationship.

The regression line predicts that a car weighting 800 kg will travel 11.7 km per litre of fuel, while a car weighing 1600 kg will travel 9.7 km per litre of fuel.

What is the slope of the least squares regression line?

- A. -400
- B. -0.0025
- C. 0.0025
- D. 400

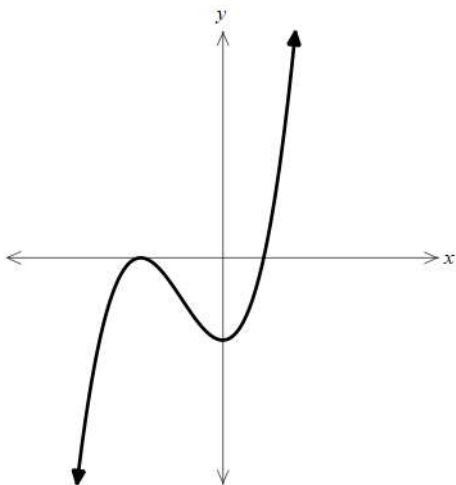
2. The domains of two continuous functions, $f(x)$ and $g(x)$, are $[-3, 2)$ and $(0, 6]$ respectively.

What is the domain of the sum function $f(x) + g(x)$?

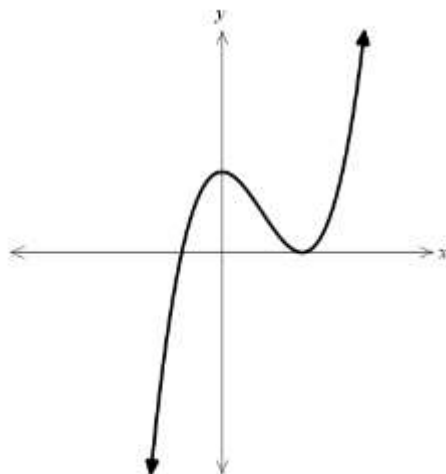
- A. $[-3, 6]$
- B. $[-3, 0) \cup (2, 6]$
- C. $[0, 2]$
- D. $(0, 2)$

3. Which of the following show the graph of $f(x) = (b-x)^2(a+x)$, if $a > 0$ and $b > 0$?

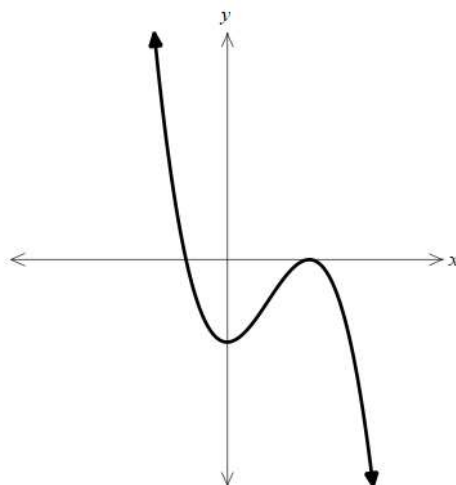
A.



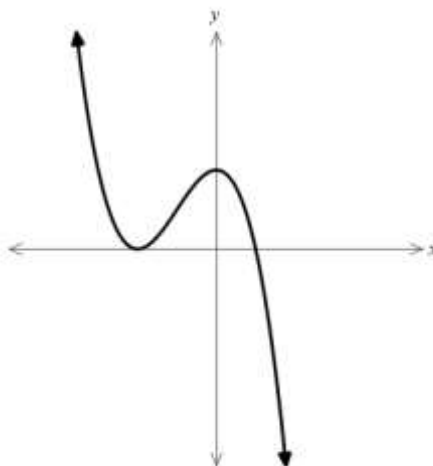
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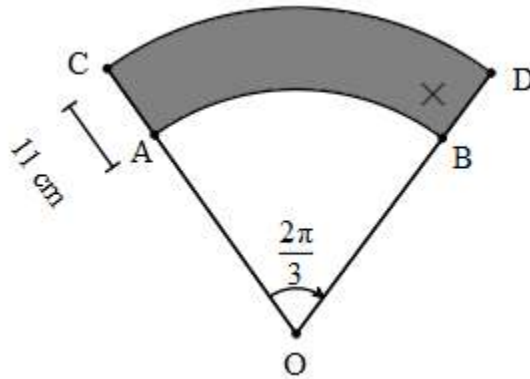
C.



D.



4. In the figure, OAB and OCD are sectors with centre O . It is given that the area of the shaded region is $143\pi \text{ cm}^2$. If $AC = 11 \text{ cm}$ and $\angle AOB = \frac{2\pi}{3}$, what is the value of OA ?



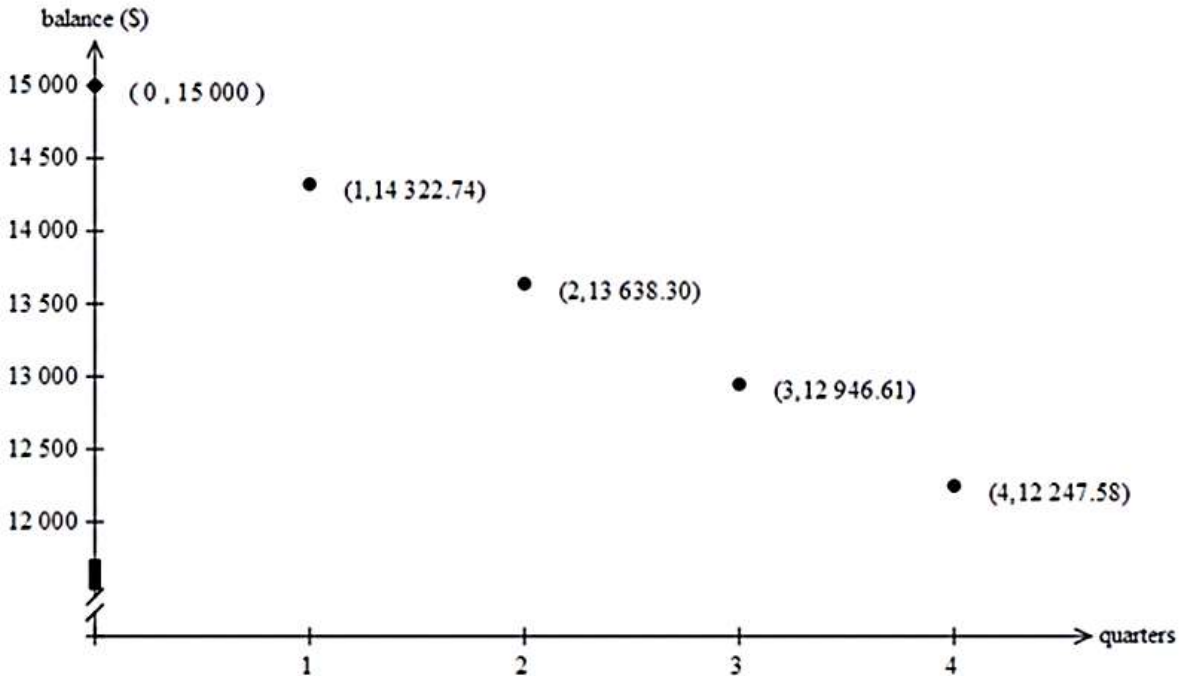
- A. 12 cm
- B. 14 cm
- C. 20 cm
- D. 25 cm

5. Let $f(x) = \ln x$ and $g(x) = \sqrt{2-x}$.

What is the domain of the derivative of $f(g(x))$?

- A. $x \leq 2$
- B. $x < 2$
- C. $0 < x < 2$
- D. $x > 0$

6. Berry took out a loan of \$15 000, with interest compounding quarterly. She makes quarterly repayments of \$836.26. The graph below represents the balance in dollars of Berry's loan at the end of each quarter for the first year of the loan.



What is the yearly interest rate for this loan?

- A. 1.05%
- B. 1.06%
- C. 4.20%
- D. 4.24%

7. A set of scores in a data set has a median of 65, an interquartile range of 16 and a variance of 5.

Each score has 2 added to it and each resulting score is then doubled to form a new set of scores.

What is the median, interquartile range, and variance of the new set of scores?

	Median	Interquartile Range	Variance
A.	130	36	10
B.	134	36	20
C.	134	32	10
D.	134	32	20

8. Given that $\int_1^6 f(x) dx = 6$ and $\int_3^6 f(x) dx = -4$. What is the value of $\int_1^3 (f(x) + x) dx$?

- A. 2
B. 6
C. 10
D. 14

9. The table below shows the probability distribution of a discrete random variable X .

X	-2	-1	0	3
$P(X = x)$	k^2	$5k$	k	$-k^2 - 6k + 1$

What is the maximum possible value for μ ?

- A. 0
- B. 1
- C. 3
- D. $\frac{589}{20}$
10. Let $f(x) = k \sin(mx + \ell)$, $m > 0$ where $f(x)$ is continuous and increasing for $a \leq x \leq b$, $f(a) = -k$, and $f(b) = k$.

Let $g(x) = k \cos(mx + \ell)$.

Which of the following statements is true for $g(x)$ in the domain $a \leq x \leq b$?

- A. $g(x)$ is always increasing
- B. $g(x)$ is always decreasing
- C. There exists a point c such that $g(c) = k$
- D. There exists a point c such that $g(c) = -k$

Trial Higher School Certificate Examination

Mathematics Advanced

Section II Answer Book

90 marks

Attempt Questions 11-34

Allow about 2 hours and 45 minutes for this section

Instructions

- Answer the questions in the space provided. These spaces provide guidance for the expected length of response.
- Your response should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet.

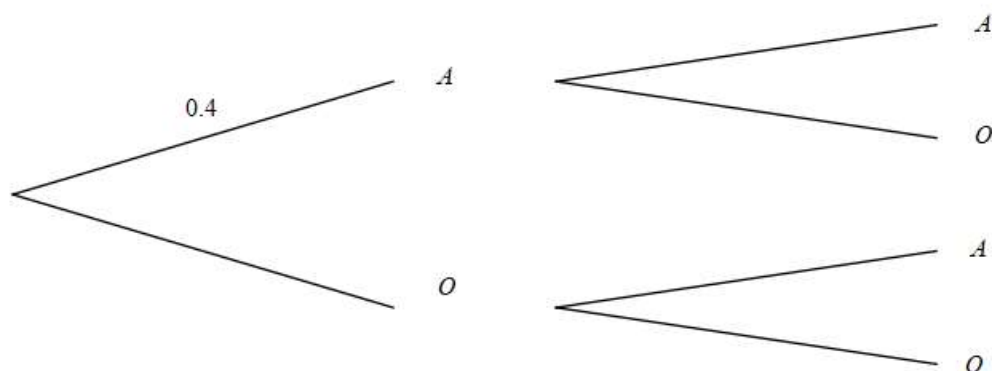
If you use this space, clearly indicate which question you are answering.

Question 11 (3 marks)

In a population 60% are of blood group O, and 40% are of blood group A.

Two people are chosen at random from this population.

- (a) Complete the tree diagram representing this situation. **1**



- (b) Find the probability that both people are of blood group A. **1**

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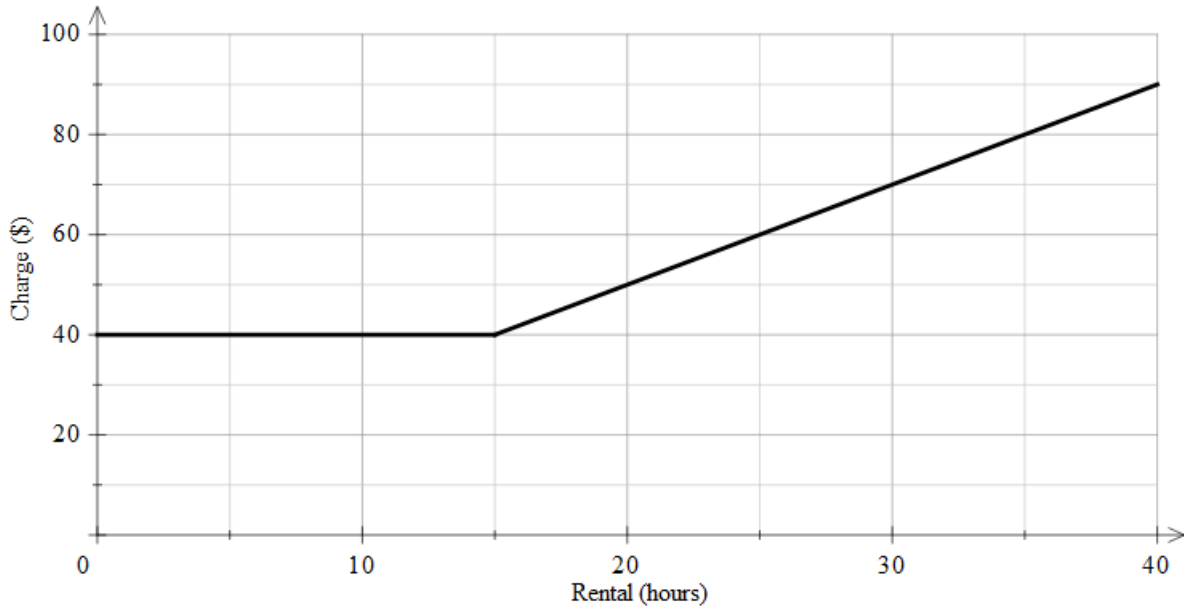
- (c) Find the probability at least one person is of blood group O. **1**

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Question 12 (3 marks)

The graph below shows the charge imposed by Company A for an electric bike. Company A charges a fixed cost of \$40 for the first 15 hours and an additional rate of \$2 per hour thereafter.



- (a) How much does the company charge for renting an electric bike for 35 hours? **1**

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- (b) Company B charges a rate of \$4 per hour, with no fixed cost. **1**

On the same grid above, draw the graph which represents Company B's pricing model.

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- (c) Hence, determine after how many hours, will Company B cost more than Company A. **1**

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Question 13 (2 marks)

On Peter's 10th birthday he started to receive an annual allowance. The first annual allowance was \$1000 and on each following birthday the allowance increased by \$500. 2

Find the total amount Peter received from his allowances up to and including his 18th birthday.

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Question 14 (2 marks)

Solve $\log_3 5 = 2 \log_3 10 - \log_3 x$. 2

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Question 15 (3 marks)

- (a) It is given that y is inversely proportional to x . **1**

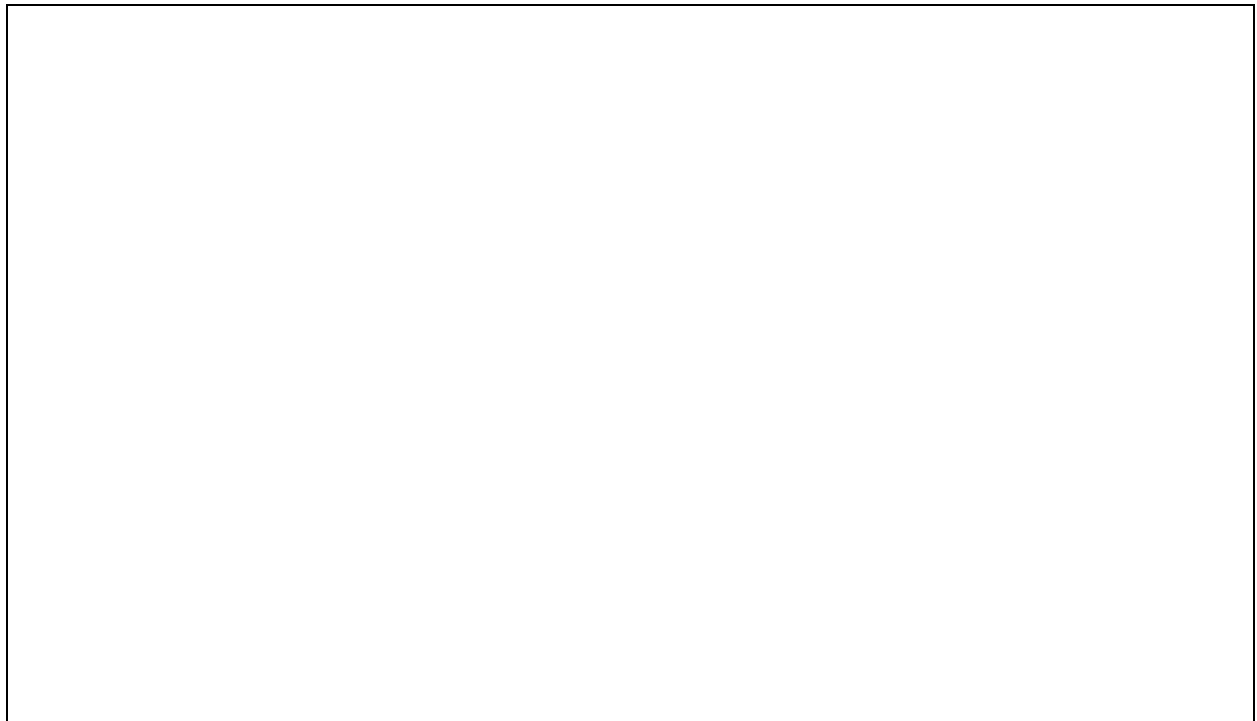
When $x = 8$, $y = 2$. Find the equation relating x in terms of y .

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- (b) If the value of x is increased from 4 to 16, find the change in the value of y . **1**

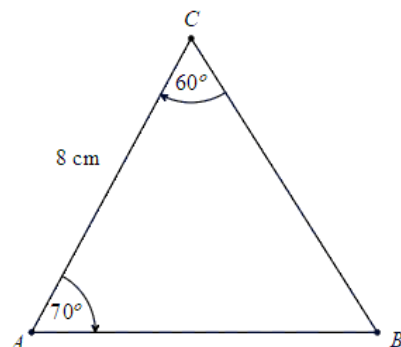
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- (c) Hence graph the relationship between x and y using the values you found previously. **1**



Question 16 (3 marks)

In the diagram below is it given that $AC = 8\text{ cm}$, $\angle C = 60^\circ$ and $\angle A = 70^\circ$.



- (a) Show that the length of $AB = 9.04\text{ cm}$.

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- (b) Hence, find the area of the triangle to 2 decimal places.

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Question 17 (3 marks)

Find the exact equation of the normal to the curve $y = \log_3 x$ at the point $x = 3$. **3**

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Question 18 (2 marks)

Given $\tan \alpha = -0.2$ and $\sin \alpha > 0$, find the exact value of $\cos \alpha$. **2**

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Question 19 (3 marks)

The table below shows the future value of a \$1 annuity at different interest rates over different numbers of time periods

Time Period	Interest rate				
	1%	2%	3%	4%	5%
1	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101
5	5.1010	5.2040	5.3091	5.4163	5.5256
6	6.1520	6.3081	6.4684	6.6330	6.8019
7	7.2135	7.4343	7.6625	7.8983	8.1420
8	8.2857	8.5830	8.8923	9.2142	9.5491

- (a) What is the value of an annuity that would provide \$315 932 after 7 years at 4% per annum compound interest. **1**

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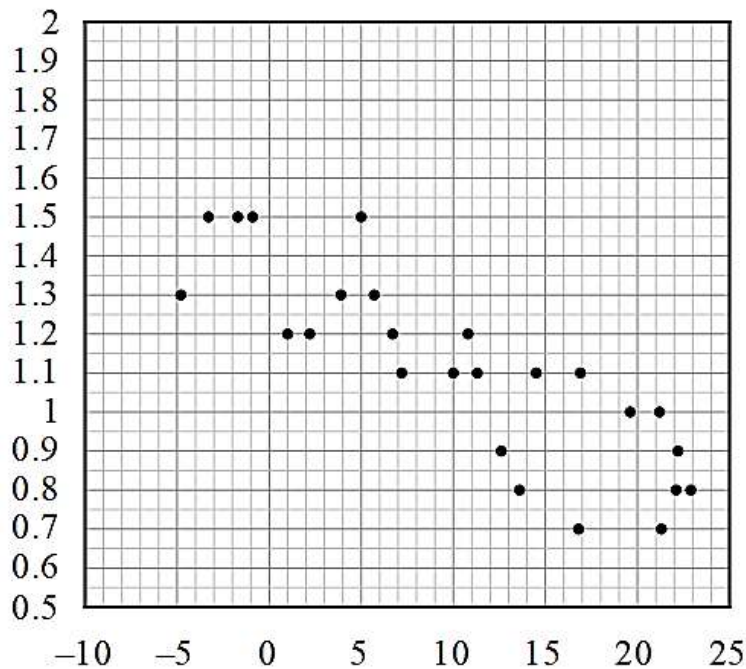
- (b) An annuity of \$5000 bi-annually is invested at 4% per annum, compounded every six months for 3 years. What is the amount of interest earned? **2**

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Question 20 (5 marks)

A researcher assumes that there is a linear relationship between the temperature and hot chocolate consumption.

Data was collected randomly in cities across Europe and the results were recorded in the scatterplot below.



The scatterplot shows the average monthly hot chocolate consumption, in litres/person against average monthly temperature, in $^{\circ}\text{C}$.

The correlation (r) is found to be -0.8492349498.

When a least squares line is fitted to the scatterplot, the equation is found to be:

$$\text{Consumption} = 1.3599 - 0.0241 \times \text{temperature}$$

- (a) Draw the least squares regression line on the scatterplot graph above.

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Question 20 continues on page 17

Question 20 (continued)

- (b) Describe the association between average hot chocolate consumption and average monthly temperature in terms of strength, direction, and form. **1**

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- (c) Interpret the slope of the least squares regression line in terms of the variables hot chocolate consumption and temperature. **1**

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- (d) Use the equation of the least squares regression line to predict the average monthly hot chocolate consumption, in litres per person, when the average monthly temperature is -6°C . **1**

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- (e) State whether your prediction in part (d) is interpolation or extrapolation. **1**

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End of Question 20

Question 21 (2 marks)

Evaluate $\int_1^3 \frac{24x^2 - 14}{4x^3 - 7x + 5}$ leaving your answer in simplest exact form. **2**

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Question 22 (3 marks)

(a) Show that $\frac{\operatorname{cosec} A}{\operatorname{cosec} A - \sin A} \equiv \sec^2 A$. **2**

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(b) Hence, or otherwise, find $\int \tan^2 x \left(\frac{\operatorname{cosec} x}{\operatorname{cosec} x - \sin x} \right) dx$. **1**

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Question 23 (4 marks)

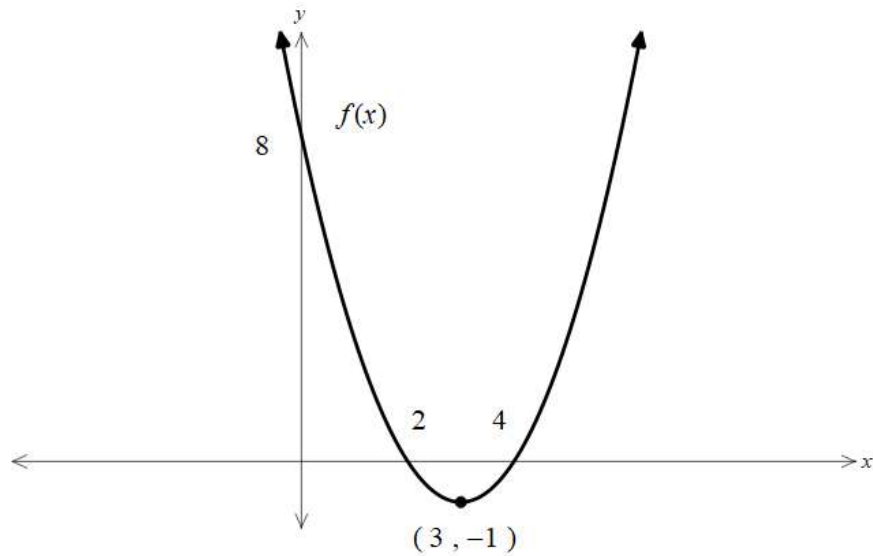
Let $f(x) = (x-3)^2 - 1$.

A transformation maps $f(x)$ to $g(x)$ such that $g(x) = \frac{1}{2} f(2-x)$.

- (a) Complete a possible sequence of transformations that maps $f(x)$ to $g(x)$. 2

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- (b) On the same set of axes sketch the graph of $g(x)$ below. 2



Question 24 (4 marks)

A muesli packet is labelled as having a net weight of 500 grams. The production process produces packets whose weight are normally distributed with a mean of 503.5 grams, and a standard deviation of 3.5 grams.

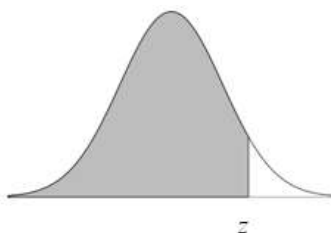
- (a) If there are 1000 packets produced, how many packets will contain less than the labelled weight? 2

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The table below gives the probability of a random variable z for some positive values of z .

z	0.80	0.81	0.82	0.83	0.84	0.85	0.86	0.87
Probability	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078

The probability values given in the table are represented by the shaded area in the following diagram.



- (b) Using this table, above what weight, would the top 20% of the packets weigh? 2

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Question 25 (4 marks)

Jenny planted a silver maple tree in her garden when it was 110 cm tall.

After one year the silver maple grew 80 cm.

Each subsequent year's growth was 90% of the previous year's growth.

- (a) How much did the silver maple tree grow in its second year? **1**

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- (b) How tall was the silver maple tree after three years? **1**

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- (c) Explain why the silver maple tree will never grow above 10 metres. **2**

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Question 26 (5 marks)

A company printer depreciates in value $V(t)$ over time t in years.

The printer is initially \$90,000 and will drop to \$0 when $t = k$.

The rate of depreciation with respect to time t is

$$V'(t) = 200(2t - 45) \quad \text{for } 0 \leq t \leq k.$$

- (a) Find $V(t)$ for $0 \leq t \leq k$. 2

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- (b) Find the value of k . 1

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- (c) Calculate the total depreciation in the first 5 years. 2

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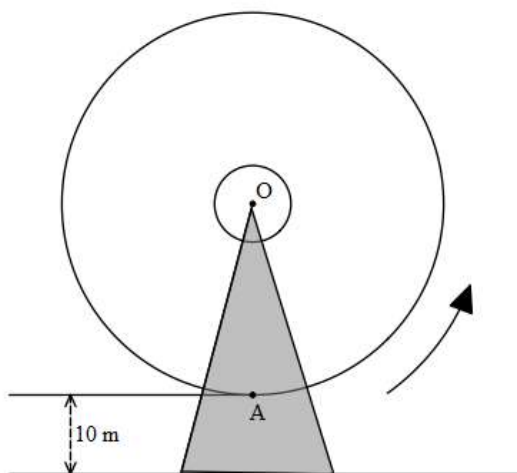
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Question 27 (6 marks)

The diagram of an observation wheel is shown below with its centre at point O. The wheel moves anticlockwise at a constant speed and completes one full rotation every 30 minutes.

When the wheel is at its lowest point (A), it is 10 metres above ground.

The radius of wheel is 50 metres.



The height above ground of a carriage pod initially at point A can be modelled by the function $h(t) = A\cos(kt) + c$, after time t minutes.

- (a) Find the values of A , k and c 2

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- (b) Find at what times the observation wheel is 85 metres above ground. 2

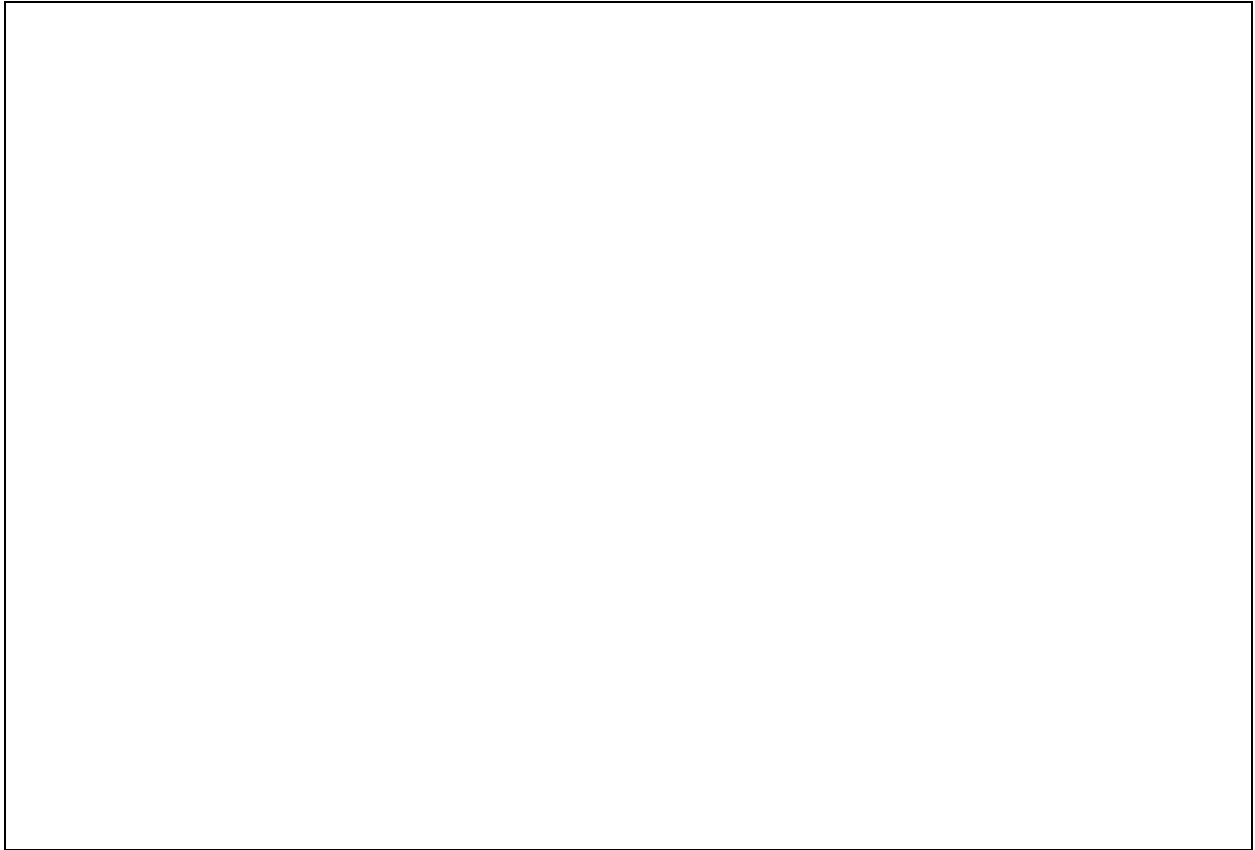
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Question 27 continues on page 24

Question 27 (continued)

(c) Sketch the graph of $h(t)$ showing two full cycles.

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End of Question 27

Question 28 (4 marks)

A polynomial function $f(x)$ has the following properties.

	$x < 0$	$x = 0$	$0 < x < 1$	$x = 1$	$1 < x < 3$	$x = 3$	$x > 3$
$f(x)$		-27		-16		0	
$f'(x)$	< 0	0	> 0	> 0	> 0	0	> 0
$f''(x)$	< 0	> 0	> 0	0	< 0	0	> 0

- (a) Find the range of values where $f(x)$ is increasing and concave up. **1**

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- (b) State the nature of the stationary points of $f(x)$. **1**

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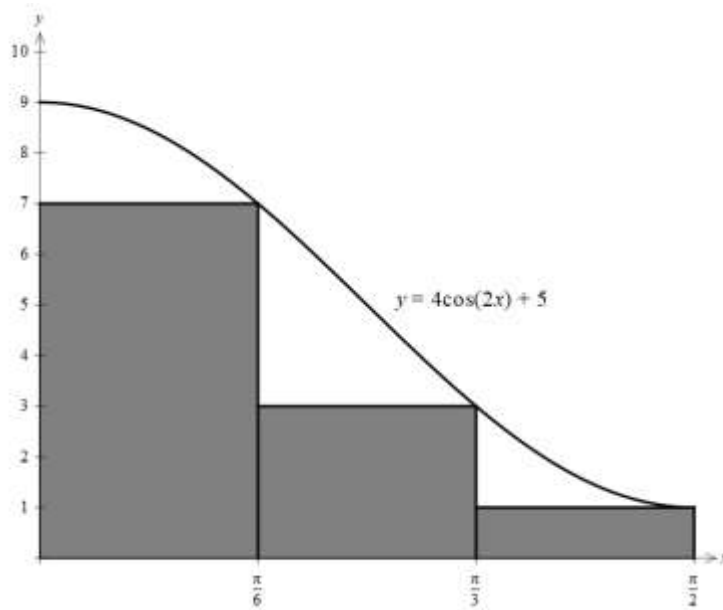
- (c) Sketch a possible graph of $y = f(x)$, using the properties in the table above. **2**

Question 29 (4 marks)

Daniel approximates the area between the x -axis, the graph of $y = 4\cos 2x + 5$

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and the lines $x = 0$ and $x = \frac{\pi}{2}$, using three rectangles as shown in the diagram below.



What is the value of Daniel's approximation as a percentage of the exact area?

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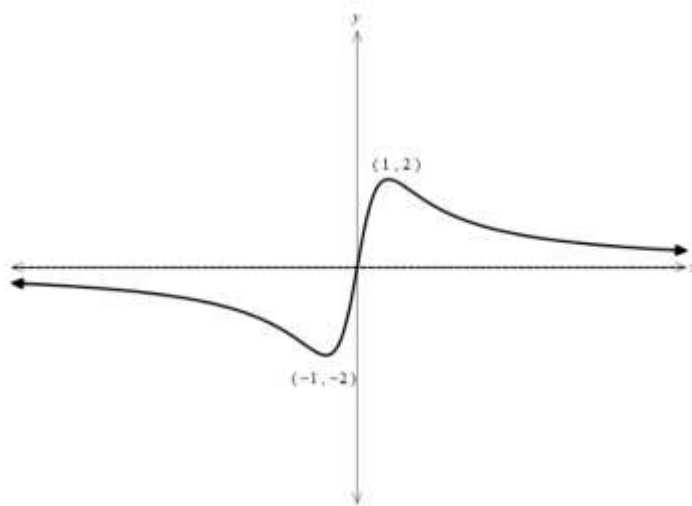
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Question 30 (3 marks)

The graph below shows $f(x) = \frac{cx}{x^2 + ax + b}$.

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There is a horizontal asymptote at $y = 0$, and a maximum and minimum stationary point at $x = 1$ and $x = -1$ respectively.

Given $f(x)$ is an odd function find the values of a , b , and c .

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Question 31 (6 marks)

The queuing time, t (in minutes), at a restaurant has a probability density function given by

$$f(t) = \begin{cases} kt(25 - t^2) & 0 \leq t \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Show that $k = \frac{4}{625}$. **1**

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- (b) Determine the mode waiting time. **2**

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Question 31 continues on page 29

Question 31 (continued)

- (c) What is the probability that a person has to queue for more than 3 minutes, given that they have already queued for 1 minute?

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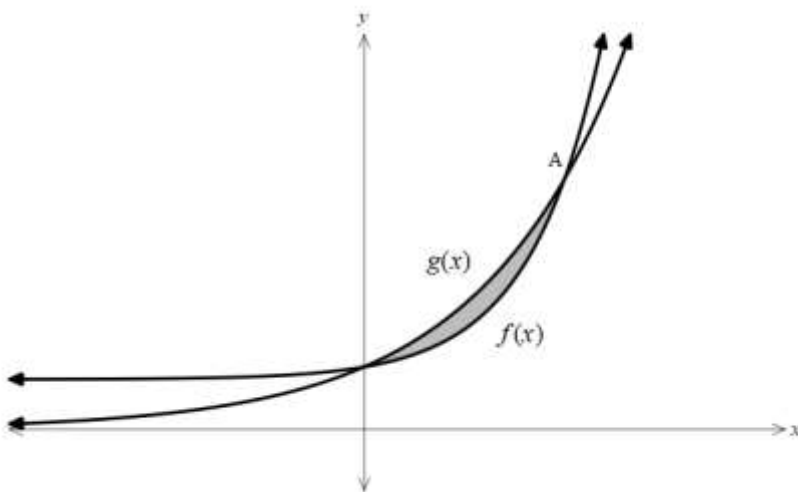
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End of Question 31

Question 32 (4 marks)

The diagram shows the graphs of $f(x) = 2^{2x} + 4$ and $g(x) = 5 \times 2^x$ below.



- (a) Show that the x -coordinate of A is 2.

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- (b) Find the exact area of the shaded region bounded by the two curves.

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Question 33 (8 marks)

In a 100m race, the speeds, S_I m/s and S_H m/s, of two animals, an iguana and a hippopotamus respectively can be modelled by the functions.

$$S_I = \frac{6400}{953125} \left(\frac{t^3}{3} - \frac{77}{4} t^2 + 325t \right), \text{ and}$$

$$S_H = \frac{2453}{1000} t e^{-0.08t},$$

Where t is the time measured from the start in seconds.

It is known that the iguana finishes the race in 12.5 seconds. [Do Not Prove This]

- (a) Use the trapezoidal rule with 5 sub-internals to estimate the distance covered by the hippopotamus in 12.5 seconds. 2

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- (b) Find $\frac{d^2 S_H}{dt^2}$ and determine whether the hippopotamus finishes ahead of the iguana. 2

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Question 33 continues on page 32

Question 33 (continued)

- (c) Differentiate $\left(\ln\left(\frac{t+2}{2}\right)\right)^2$. 2

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In the same race, the speed S_c m/s, of a cow is modelled by

$$S_c = \frac{50[\ln(t+2) - \ln 2]}{t+2}.$$

- (d) Determine the finishing position of all 3 animals. 2

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End of Question 33

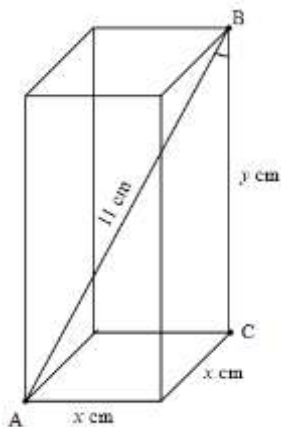
Question 34 (4 marks)

The diagram below shows the frame of a rectangular prism made with iron wire.

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It has a square base of side x cm and a height of y cm. The diagonal AB is 11 cm.

The total length of wire used for the frame (including the diagonal AB) is 71 cm.



Find the maximum value of $\angle ABC$ to the nearest degree.

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End of Exam

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Section II extra writing space

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